Regime-Sensitive Performance Analysis of Machine Learning Models for Stock Prediction

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Abstract

This study looks at how well machine learning models work for predicting stock returns in different market situations.

Using daily data from the SPY index over the years 2000 to 2026, four models—Linear Regression, Ridge Regression, Random Forest, and LSTM—are tested using a walk-forward validation method. The models are checked using statistical measures and trading-based performance indicators like the Sharpe ratio and information coefficient.

The results show that although these models have about 55-59% accuracy in predicting the direction of stock movements and have some non-zero information coefficients, their performance after considering risk is not very strong.

The best model, Ridge Regression, has a Sharpe ratio of 0.29, which is actually worse than a random guess, which has a Sharpe ratio of 0.76. These findings suggest that the signals used for prediction are not strong enough to be used for making money.

Overall, the results support the idea that highly liquid equity markets are efficient in the weak-form sense, meaning that past prices don't give a clear advantage in predicting future prices.

I. Introduction

Financial markets are a big area of study under the Efficient Market Hypothesis, which states that asset prices include all available information, making them hard to predict.

Despite this, there is growing interest in using machine learning techniques on financial time series to find hidden patterns that can help in predicting stock behavior.

Financial markets often show different behaviors based on underlying factors like trends and volatility.

These different conditions, known as market regimes, can have a big impact on how well a model performs. Traditional methods usually look at overall metrics, which may not show how models perform in different regimes.

This study checks whether machine learning models can consistently predict stock returns across various market conditions.

Instead of only looking at accuracy, the study focuses on how stable and meaningful the model outputs are. By using a realistic trading simulation and comparing model performance to a random benchmark, the study tries to see if the predictions can lead to real financial benefits.

II. Literature Review

The use of machine learning in financial markets has been widely studied.

Fama's work in 1970 laid the foundation for the Efficient Market Hypothesis, which sets the stage for understanding the predictability of markets. Later studies have tried to challenge this idea with statistical and machine learning methods.

Linear models such as regression are popular because they are easy to understand and use.

Ridge regression builds on this by adding regularization to prevent overfitting. Ensemble methods like Random Forest, introduced by Breiman, have shown strong performance in handling complex nonlinear relationships across different fields.

Deep learning, especially LSTM networks, has gained attention for its ability to model time-dependent sequences.

However, financial data comes with its own set of challenges, such as noise, non-stationarity, and low signal-to-noise ratios.

Many studies point out that even though machine learning models can offer moderate predictive accuracy, turning those predictions into profitable trading strategies is still tough.

Factors like transaction costs, market efficiency, and overfitting often limit how useful these models are in real-world settings. This study adds to the conversation by including regime-based evaluation and realistic backtesting.

III. Problem Statement

The main goal of this study is to see if machine learning models can reliably and meaningfully predict stock returns in different market conditions.

While many models are good at statistical accuracy, it is not clear whether these predictions can be used for profitable trading in real life.

This study aims to close the gap between predictive modeling and real-world trading by using regime-based analysis, walk-forward validation, and realistic trading simulations.

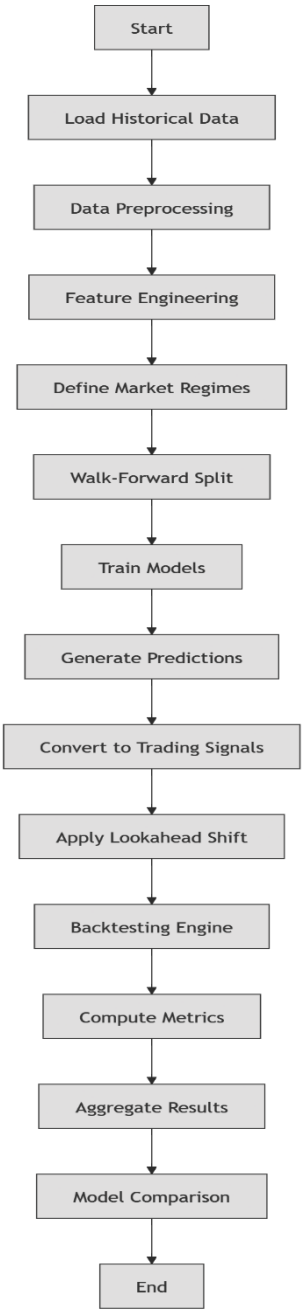
The focus is on measuring robustness, stability, and risk-adjusted performance rather than just the accuracy of the predictions.

IV. System Architecture

A. Flow Description

The system follows a structured process that starts with gathering and preparing data, then moving on to creating features and identifying market conditions.

Once the data is ready, it is used to train various machine learning models. These models generate predictions, which are then turned into trading signals. These signals are tested using a backtesting tool.Finally, the performance of the system is measured using statistical and financial measures.



B. Algorithm

The overall process starts by collecting historical stock prices and cleaning and normalizing the data.

Then, features are created by looking at past returns and calculating rolling statistics like averages and volatility. Market conditions are identified by analyzing trends and volatility.

The data is split into training and testing sets using a time-based approach.

Models are trained on the training set and tested on the future data. Predictions from each model are converted into trading signals based on whether they are positive or negative. Small signals are filtered out using a threshold. These signals are shifted forward to avoid using future data incorrectly. The trading strategy is then simulated, taking into account transaction costs.

Performance measures such as RMSE, directional accuracy, Sharpe ratio, and information coefficient are calculated to assess how well the models work.

Input: Historical stock price data D

Output: Model performance metrics and trading results

Step 1: Load historical price data for the selected stock and time period.

Step 2: Clean the data by removing missing values and calculate daily returns.

Step 3: Generate features like lagged returns and rolling statistics such as average and volatility.

Step 4: Define market regimes by using moving averages (short and long-term) to detect trends and rolling standard deviation to measure volatility.

Step 5: Split the data using walk-forward validation by creating sequential training and testing windows.

Step 6: For each model (Linear Regression, Ridge Regression, Random Forest, LSTM), train the model on the training data and make predictions on the test data.

Step 7: Convert model predictions into trading signals—positive predictions mean a long position, negative predictions mean a short position, and small signals are filtered.

Step 8: Adjust signals to prevent using future data incorrectly by shifting them forward by one period.

Step 9: Simulate the trading strategy by calculating returns based on the signals and subtracting transaction costs.

Step 10: Calculate performance metrics including RMSE, directional accuracy, Sharpe ratio, and information coefficient.

Step 11: Combine results from different market regimes.

Step 12: Compare model performance and rank them based on their results.

Step 13: Output the final results along with visual representations.

V. Methodology

A. Data

The analysis uses historical daily price data for the SPY index from January 2000 to January 2026.

The data is obtained via an external API and is preprocessed, including cleaning and creating new features. Missing data and anomalies are removed before the models are trained.

B. Models

The study looks at four models—Linear Regression, Ridge Regression, Random Forest, and LSTM.

These models range from simple linear methods to more complex nonlinear and deep learning techniques, allowing for a comparison of how performance changes with model complexity.

C. Feature Engineering

Features are created from historical price data and include lagged returns and rolling statistics.

Care is taken to ensure that only past data is used in feature creation to avoid looking ahead. Market regimes are defined based on trends and volatility. Trends are identified using moving average relationships, while volatility is calculated from rolling return variance. This setup helps in testing how models perform under different market conditions.

D. Validation Framework

A walk-forward validation approach is used where models are trained on a growing window of data and evaluated on the next set of unseen data.

This mimics real-world use and helps prevent overfitting that can happen with fixed data splits.

E. Trading Strategy

Model predictions are turned into trading signals by assigning long positions when predictions are positive and short positions when they are negative.

Weak signals are filtered out using a small threshold. Positions are shifted forward by one period to avoid looking ahead, and transaction costs are included to simulate real trading conditions.

F. Evaluation Metrics

The models are judged using a mix of statistical and financial measures, including RMSE, directional accuracy, Sharpe ratio, information coefficient, cumulative return, and regime dispersion.

These metrics give a comprehensive view of both how well the models predict and whether their predictions are economically useful.

VI. Results and Discussions

A. Predictive Performance

Ridge Regression and Linear Regression had similar RMSE values, showing similar predictive accuracy.

Random Forest performed better in directional accuracy, suggesting it predicted the direction of returns better. LSTM had higher error rates and lower stability, indicating challenges in capturing consistent patterns over time.

B. Trading Performance

Random Forest had the highest Sharpe ratio among the models, showing better risk-adjusted performance.

Ridge Regression performed best overall, due to a good balance across multiple metrics. LSTM had negative Sharpe values, suggesting poor trading performance despite being more complex.

C. Visualization and Diagnostics

To understand how models behave under different conditions, visualizations were created.

As shown in Figure 1, performance varies significantly across different market regimes, with unstable information coefficients.

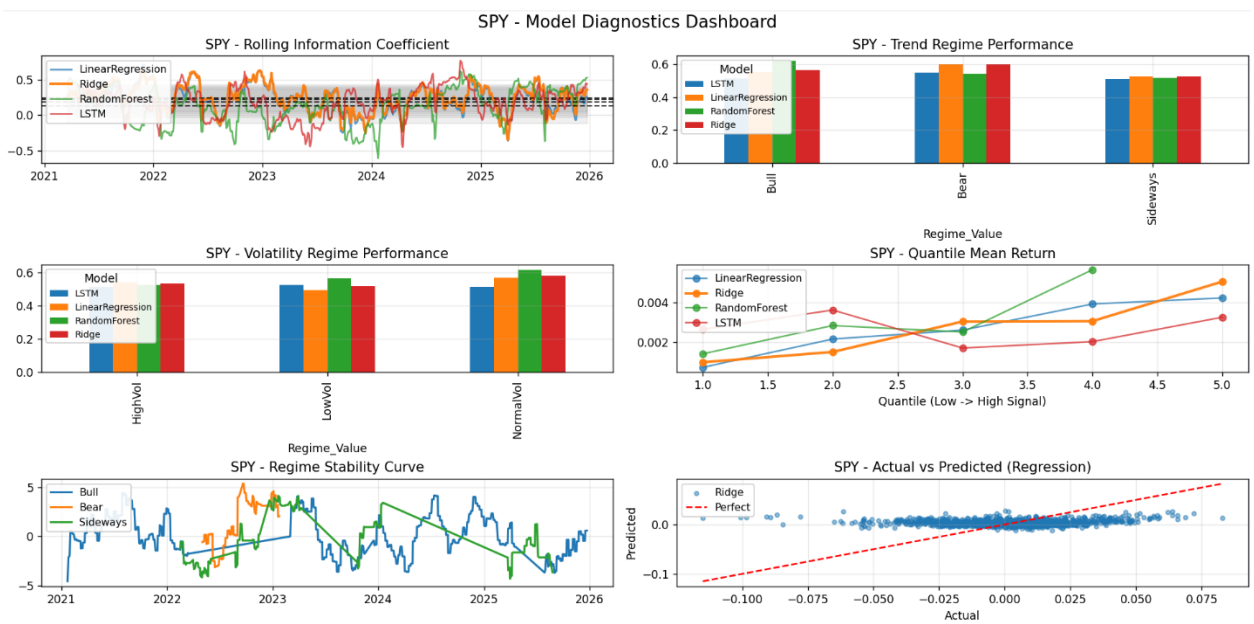


Figure 1: Model diagnostics showing rolling information coefficient, performance across regimes, and how well predictions match actual outcomes..

The performance of the models is illustrated in Figure 2.

Equity curves and rolling Sharpe ratios show that the models are not consistently profitable.

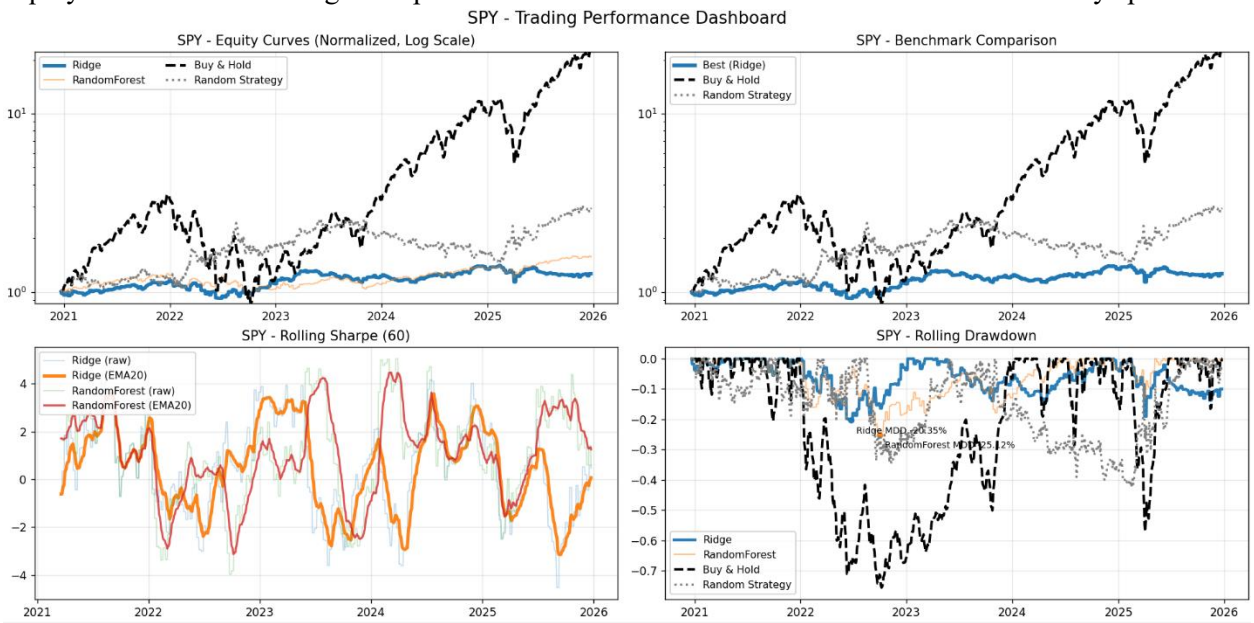


Figure 2: Trading performance comparison using equity curves, rolling Sharpe ratios, and drawdowns across models.

D. Information Coefficient

Ridge and Linear Regression models showed moderate information coefficients, indicating some correlation between predictions and actual returns.

Random Forest had lower correlation, and LSTM showed inconsistent results.

E. Market Efficiency Comparison

The best model achieved a Sharpe ratio of around 0.29, while a random strategy had a higher ratio of approximately 0.76.

This suggests that no model consistently outperforms a random approach when considering risk.

F. Discussion

The results suggest that while machine learning models can pick up on some predictive signals, these signals are not strong enough to generate consistent profits after accounting for noise and transaction costs.

Simpler models like linear ones performed as well as more complex models, and increased complexity didn't always improve performance. Predictive signals tend to weaken in a trading environment, and variations between market states aren't strong enough to offer a reliable advantage. These findings support the idea of weak-form market efficiency.

VII. Conclusion

This study tested four machine learning models within a regime-sensitive framework using walk-forward validation and realistic trading simulations.

Although the models showed some predictive ability, none consistently outperformed a random strategy. The findings support the idea that there are limited exploitable patterns in liquid equity markets.

VIII. Future Scope

Future work could look into other features like macroeconomic indicators and sentiment data, which might provide more predictive insights.

Using ensemble techniques that combine multiple models could increase reliability. Additionally, exploring higher-frequency data and different modeling approaches might help better understand short-term market movements.

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